

Overview Task 10.3

For Task 10.3HD, I implemented the F1 Strategy Lab, an interactive race strategy simulator within my Formula 1 World website. It develops the feature proposed in Task 7.3HD into a working application. Formula 1 fans can use it to explore how their strategy choices affect a hypothetical race, with results they can save and compare.

Users choose a season and race, then select a driver. They configure tyre stints and pit stops and adjust assumptions such as base lap time, fuel effect, traffic loss and pit-stop loss before running the simulation. The results show total race time and average lap time, along with the number of pit stops and individual lap data. Saved strategies and previous runs remain available for comparison.

The simulation uses the Web Workers API to run calculations separately from the browser's main thread. This keeps the interface responsive during a run. Users can follow its progress or cancel it without waiting for all the calculations to finish.

Historical Formula 1 data is stored in SQLite. F1DB supplies data for the 2024 and 2025 seasons alongside the existing Jolpica and OpenF1 integrations. Together, these sources provide information about races and drivers, including constructors and historical results. The interface separates this historical context from the simulator's assumptions. A historical fastest lap, for example, changes the base lap time only when the user chooses to apply it.

The Strategy Lab connects the interface and Web Worker to server-side APIs and persistent SQLite storage. These components extend the existing website with a working simulation and stored comparison data. The following sections explain their implementation and how the feature supports usability, accessibility and performance, as required by Task 10.3HD.